

AYUSH MISHRA

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Education & Coursework

Bhagwan Parshuram Institute of Technology, GGSIPU Delhi

2022 – 2026

B.Tech in ECE (Minor in CS), CGPA: 8.0 | Coursework: DSA, Machine Learning, Computer Networks, Mobile Computing

Projects

Assessly - AI-Proctored Mock Interview Platform [Live](#) | [Github](#)

2026

React 19, Node.js, Express, MongoDB, MediaPipe, Deepgram, Docker, AWS EC2

- Architected a resume-grounded interview engine using **DeepSeek V3 (OpenRouter)** that generates personalized technical interviews and dynamically injects follow-up questions based on candidate responses.
- Built a low-latency voice interview pipeline streaming audio to **Deepgram STT** over WebSockets with ephemeral server-minted tokens, paired with neural **edge-tts** for the AI interviewer's voice.
- Built a real-time client-side **MediaPipe** proctoring system, converting noisy per-frame detections into stable signals via debounced thresholds, cooldowns, a calibration baseline, and GPU-to-CPU fallback, enforced by a tamper-proof server-authoritative strike state machine using atomic MongoDB transitions.
- Secured and deployed a production MERN stack on **AWS EC2** with **Docker**, **Nginx**, and automated **HTTPS**, hardened with **Firebase** token verification, **JWT** cookies, per-endpoint IDOR ownership checks, **Razorpay HMAC** verification, and per-user rate limiting.

CodeSync - Multiplayer IDE with AI Pair Programming [Live](#) | [Github](#)

2026

React, Node.js, Socket.io, Yjs, Redis, Docker, AWS EC2, Gemini API

- Engineered conflict-free collaborative editing using **Yjs CRDTs** over **Socket.io WebSockets**, enabling sub-second synchronization of code, cursors, and user presence across multiple clients.
- Architected **Redis-backed** room persistence, Yjs document restoration, and collaborative session recovery across reconnects and server restarts.
- **Self-hosted** Piston in isolated **Docker** containers to securely execute code across **12 programming languages** with sandboxed runtime environments and stdin support.
- Integrated token-streamed Gemini responses into shared CRDT state for collaborative debugging and AI-assisted pair programming; **deployed the stack on AWS EC2 using Docker Compose and automated HTTPS**.

Banking Ledger [Github](#)

2026

Node.js, Express 5, MongoDB, Mongoose, JWT

- Engineered a double-entry ledger system using **MongoDB aggregation pipelines** for balance computation, eliminating stored-state inconsistencies.
- Enforced **idempotent transfers** and **ACID-compliant** multi-document transactions to prevent double-spending under concurrent workloads.
- Built **stateless JWT authentication**, **TTL-based token blacklisting**, append-only audit trails, and transactional **email notifications using Nodemailer**.

Experience

Software Development Intern

Aug 2024 – Nov 2024

American Business Solutions Inc. (Remote)

Ohio, USA

- Built responsive role-based dashboards using React, Material-UI, Redux, and React Hook Form for task, attendance, and authentication workflows, optimizing state management and real-time data handling to improve consistency and UX across employee/admin systems.

Web Development Head

Aug 2024 – Aug 2025

Google Developer Student Clubs, BPIT

Delhi, India

- Led web initiatives and mentored juniors in UI/UX, responsive design, and scalable React component architecture; organized technical workshops including a Snapchat AR session with 150+ participants.

Game Development Head

Jun 2023 – Aug 2024

Google Developer Student Clubs, BPIT

Delhi, India

- Led 15+ technical workshops and organized SolveSphere Challenge for 1000+ participants (80% YoY engagement growth); coordinated with 70+ GDSC chapters across NCR and hosted Google Cloud & GenAI workshops with Newton School.

Technical Skills

- **Languages:** C++, JavaScript, SQL
- **Frameworks/Libraries:** React.js, Redux, Node.js, Express.js, Mongoose, Tailwind CSS, Framer Motion
- **Databases & Tools:** MongoDB, MySQL, Redis, Git, GitHub, Docker, AWS EC2, Vite, JWT, Nodemailer, Caddy, Unity, REST APIs

Achievements

- **Global Rank 1665/39,046 (Top 4.2%)** in LeetCode Weekly Contest 495; max rating **1788 (Top ~8%)**; solved **400+** DSA problems.
- **Top 10/300+ Teams**, CodeClash Hackathon 2023, BVCOE Delhi | **Top 10/500+ Teams (Solo)**, Vihaan Hackathon 2024, DTU Delhi.